RUBA SHIHADA

Computer Systems Engineer

Gaza, Palestine | rubashihada5@gmail.com

LinkedIn: Linkedin | Portfolio: Portfolio

PROFESSIONAL SUMMARY

Computer Systems Engineer (B.Sc., First Class Honors) from Gaza building AI systems that expand access to education for students in conflict zones. Designed and developed an AI agent platform that indexed 49 fully funded scholarships for Palestinian students. Delivered a citizen-services Flutter application to the Municipality of Gaza. Seeking a fully funded Master's in Computer Science (AI) to research how LLM-based multi-agent systems can widen education access in conflict zones.

EDUCATION

Al-Azhar University – Gaza, Palestine | (2019 – 2024)
Bachelor of Engineering in Computer Systems Engineering
GPA: 90.02% *(First Class Honors)*

Al-Falluja Secondary Girls School – Gaza, Palestine | (2016 – 2019)
High School Certificate (Scientific Stream)
GPA: 96.7%

PROJECTS

Gaza & Palestine Scholarship Finder — AI Agent | (Aug 2026)

Python, Flask, Flask-SocketIO, SQLite, SQL,Cosine Similarity, OpenRouter API (LLMs), HTML/CSS

- Designed and developed an interactive web platform that helps Palestinian students locate fully funded scholarships and emergency academic grants through a smart split-screen interface and an integrated AI assistant.
- Engineered a multi-agent system led by an Orchestrator Agent, coordinating specialized experts: a Text-to-SQL DB Read Expert, a Vector Search Expert performing semantic search over 1536-dimensional embeddings, and a DB Write Expert.
- Built a responsive split-screen UI connecting a live scholarship directory and saved bookmarks with a persistent, real-time AI assistant for instant scholarship matching.

Citizen Services Mobile App | Municipality of Gaza | (Dec 2025 – Apr 2026)

Flutter, Dart, UI/UX Design, Android Studio

- Designed and developed a cross-platform mobile application for citizen services in collaboration with a team, building user-friendly interfaces using Flutter.
- Built complaint-management features that organize citizens' reports about essential services, improving response times for municipal engineers.
- Delivered a real product to a government client — supporting local infrastructure.

Pharmacy Pulse — Smart Medication Tracking System | Graduation Project | (Dec 2024)

PHP, MySQL, HTML/CSS, XAMPP, phpMyAdmin

- Developed a website that helps patients locate and compare medications across various local pharmacies.
- Integrated real-time medication price display and availability verification across different pharmacy locations.
- Built the responsive, mobile-friendly front end using HTML, CSS with PHP and MySQL powering server-side operations and data storage.

VOLUNTEER & COMMUNITY LEADERSHIP

English Tutor – Volunteer | (Aug 2026 – Present)
Paper Airplanes – California, USA (Remote)

- Conduct one-on-one virtual English tutoring sessions using curriculum provided by the organization to support a conflict-affected student.
- Adapt learning materials to match the student's learning pace, focusing on improving language proficiency and communication skills.
- Provide personalized feedback and continuous guidance to build the student's confidence in English speaking, listening, and grammar.

Flutter Developer - Volunteer | (Dec 2025 – Apr 2026)
Municipality of Gaza - Palestine

- Volunteered as part of a development team to deliver a citizen-services mobile application to the Municipality of Gaza.
- Contributed to streamlining how citizens' complaints reach municipal engineers, improving public-service responsiveness.

Peer Tutor – Volunteer | (Year – Year)

- Provided one-on-one tutoring for four high school students in mathematics, science, English, and other core subjects.
- Prepared for each session in advance by reviewing the students' school curriculum and adapting lessons to their level.
- Supported all students to graduate with excellent grades and enter their preferred university programs.

LANGUAGE SKILLS

- **Arabic** Native
- **English** B2 (Upper-Intermediate Certificate from *Paper Airplanes*, 2026)

TECHNICAL SKILLS

Programming Languages:	Python, PHP, Dart, Java, C++
AI & LLM:	Multi-Agent Architecture, LLM Orchestration, Prompt Engineering (Few-Shot, Role Prompting, Chain of Thought), Text-to-SQL, Semantic Search, Vector Embeddings, Cosine Similarity, LLM API Integration (OpenRouter — GPT-4o / Gemma)
Web & Mobile Development:	Flutter, Flask, Flask-SocketIO, Jinja2, HTML/CSS
Databases:	MySQL, SQLite, SQL
Design:	UI/UX Design, Mobile Interface Design, Graphic Design, Figma
Tools:	Git, GitHub, VS Code, XAMPP, phpMyAdmin, Postman, Android Studio
Soft Skills:	Problem Solving, Teamwork, Time Management, Critical Thinking

REFERENCES

Available Upon Request

PORTFOLIO

GitHub: GitHub
Behance: Behance